

PH3103 Mathematical Methods of Physics
Autumn Semester - 2024
Indian Institute of Science Education and Research, Kolkata
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Homework: 8

Submission Date: xx/yy/zz

1. Check whether the following sets of elements form a Linear Vector Space (LVS) over the field of real numbers or the field of complex numbers, as the case may be. If they do, find the dimensionality of the LVS.
 - (a) The set of all $(n \times n)$ matrices with complex numbers.
 - (b) The set of all real solutions of the differential equation $y'' - 3y' = 2y = 0$, where prime denotes derivative of the function $y(x)$.
 - (c) The set of all $(n \times n)$ unitary matrices.
2. Let $|\phi\rangle$ and $|\psi\rangle$ are two linearly independent vectors in a real LVS. Find the value of real scalar α that makes $\|\psi - \alpha\phi\|$ a minimum, and find this minimum value.
3. Use the Cauchy-Schwarz inequality to establish triangle inequality.
4. Consider a general (2×2) matrix M with elements of complex numbers a, b, c, d .
 - (a) Treating this matrix as an element of a LVS of 2×2 matrices, make an expansion of M in terms of four (2×2) matrix with each one with one non-zero element of value 1. This is the 'natural' expansion (basis) of a general (2×2) matrix M .
 - (b) Now, expand M in terms of three Pauli matrices and one unit matrix. Show that the expansion coefficients can be *uniquely* written in terms of the expansion coefficients in the natural basis. This shows that three Pauli matrices and one unit matrix form a basis in the LVS of all (2×2) matrices.
 - (c) Make an observation in the expansion in terms of Pauli matrices and the unit matrix, and argue that any traceless (2×2) matrix can be expanded uniquely as a linear combination of the three Pauli matrices.
5. Let $\vec{a}$ is an ordinary three vector. Then show the following identity

$$e^{i\vec{a} \cdot \vec{\sigma}} = I \cos(a) + i \frac{(\vec{a} \cdot \vec{\sigma})}{a} \sin(a), \quad (1)$$

where a is norm of the vector $\vec{a}$. The exponential of a matrix has the usual meaning in terms of power series expansion.

6. Show that the following definition $(A, B) = \text{Tr}(A^\dagger B)$, where A and B are two $(n \times n)$ matrices satisfies the properties of scalar/inner product. Here $A^\dagger$ is the hermitian conjugate of A .
7. In a two-dimensional LVS, let $|\phi_1\rangle, |\phi_2\rangle$ constitute an orthonormal basis. Consider the operator $H = a(|\phi_1\rangle\langle\phi_1| - |\phi_2\rangle\langle\phi_2| + |\phi_2\rangle\langle\phi_1| + |\phi_1\rangle\langle\phi_2|)$ with a is a real constant. Find the eigenvalues and eigenvectors of H , and express the later as a linear combinations of $|\phi_1\rangle$ and $|\phi_2\rangle$.
8. Identify the function $f(x)$ that belongs to $L_2(-\infty, +\infty)$: (a) $(x^2 + 1)^{-1/2}$ (b) $\sin x/x$ (c) e^{-1/x^2} (d) $(x^2 - 1)^{-1}$

9. Using the Cauchy-Schwarz inequality in $L_2(-\infty, +\infty)$ show that

$$\int_{-\infty}^{\infty} \frac{e^{-x^2/2} dx}{(x^2 + 1)^{1/2}} < \pi^{3/4} . \quad (2)$$

Using numerical integration to understand how well is the limit.

10. Starting from the linearly independent set $1, x, x^2, \dots$ on $L^2[-1, 1]$, obtain the first three orthonormal polynomials.
11. Plot the first few terms of the following infinite sequence of continuous functions defined in the interval $[-1, +1]$

$$f_k(x) = \begin{cases} 1 & \text{if } 1/k \leq x \leq 1; \\ (kx + 1)/2 & \text{if } -1/k \leq x \leq 1/k; \\ 0 & \text{if } -1 \leq x \leq -1/k. \end{cases}$$

Argue that it is a Cauchy sequence, but not convergent. Therefore we have ONE (and one is enough) sequence of continuous functions that is not convergent. This proves that the space $C[-1, 1]$ is incomplete, and therefore can not form a Hilbert space.